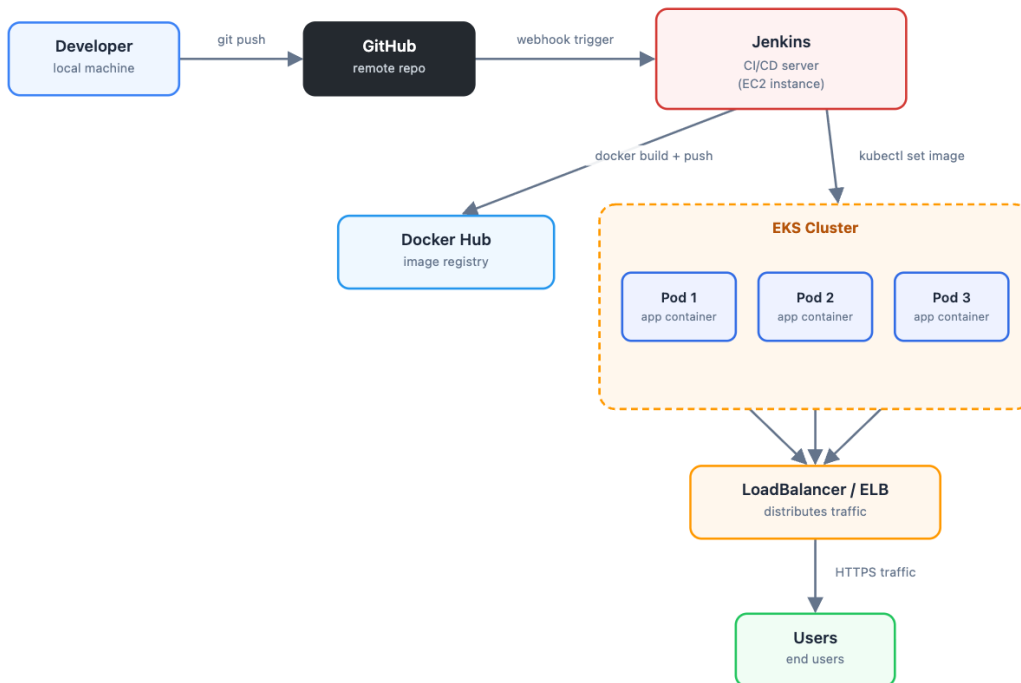


Assignment 2 - Lizon Raj, SWE 645

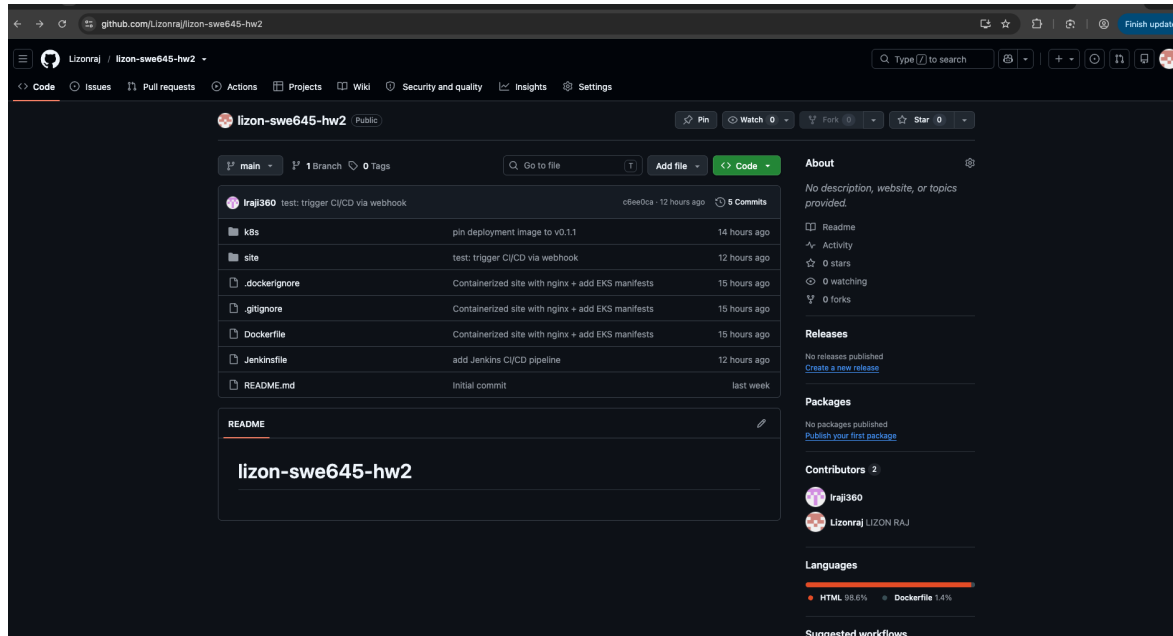
This document captures the deliverables for SWE 645 Homework Assignment 2 — containerizing the Assignment 1 static site, deploying it on Amazon EKS with 3 self-healing replicas behind a LoadBalancer, and wiring a Jenkins CI/CD pipeline triggered by GitHub webhooks. Each section below shows the corresponding artifact or proof of a required behavior.

Pipeline Architecture and Flow:



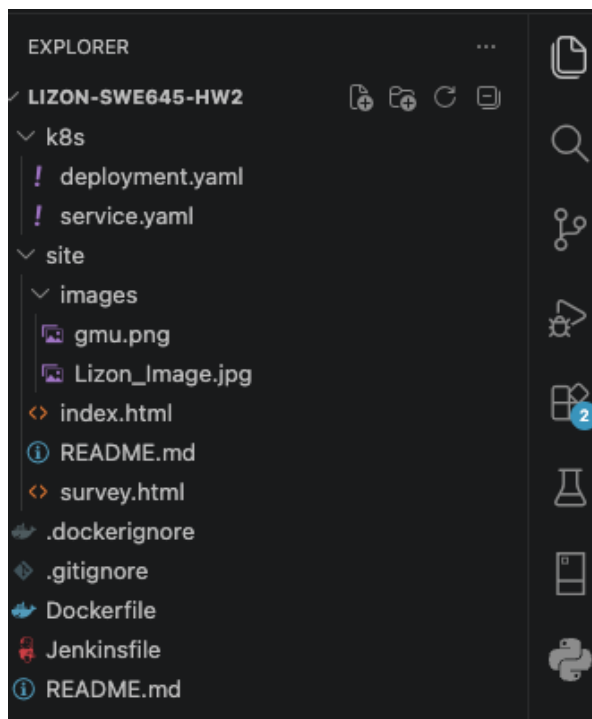
GitHub Repo:

Public GitHub repository containing all source code, Kubernetes manifests, Dockerfile, Jenkinsfile, and documentation for the assignment. The webhook configured on this repository triggers the CI/CD pipeline on every push to main.



Repo Tree:

Repository layout at the root. site/ holds the static HTML from Assignment 1; k8s/ contains the Deployment and Service manifests; Dockerfile and .dockerignore define the container image; Jenkinsfile defines the CI/CD pipeline; docs/ holds this documentation.



Docker File:

Multi-stage-free Dockerfile using nginx:1.27-alpine as the base image (~23 MB) and copying the entire site/ directory into nginx's default web root (/usr/share/nginx/html/). Chose alpine for small footprint and nginx for a production-grade static file server.

```
Dockerfile x
Dockerfile > ...
1 # Dockerfile - packages the student application into a container. Written by Lizon Raj for SWE 645. Assignment 2.
2
3 FROM nginx:1.27-alpine
4
5 COPY site/ /usr/share/nginx/html/
6
7 EXPOSE 80
```

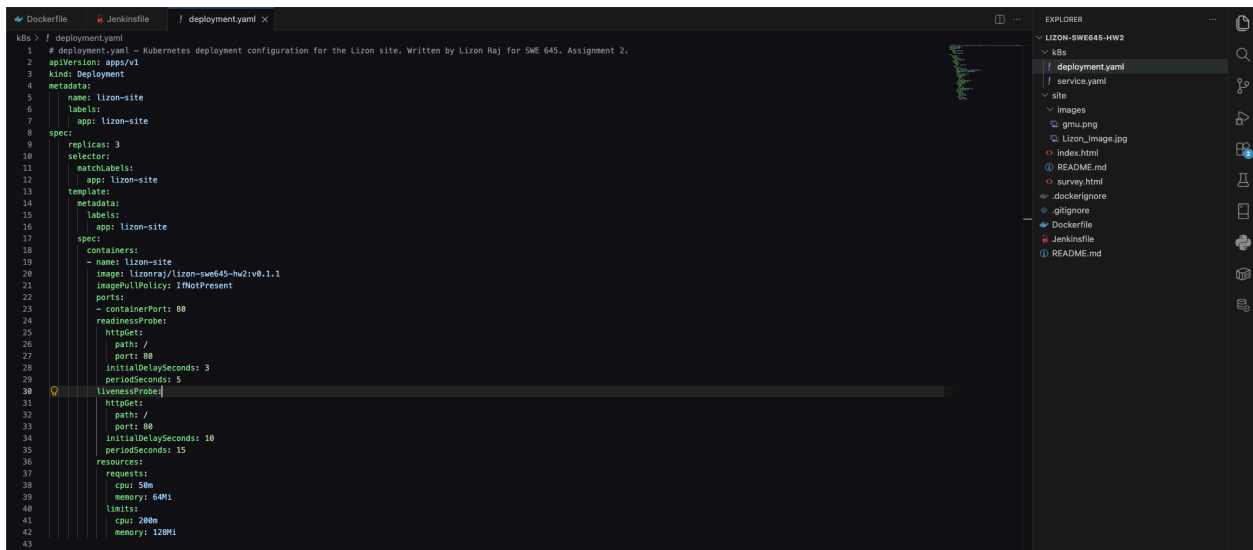
Jenkins File:

Declarative Jenkins pipeline with five stages: Checkout → Compute tag → Docker build → Docker push → Deploy to EKS. The pipeline captures the short git SHA as an immutable image tag, pushes both the SHA tag and :latest, then triggers a rolling deploy via kubectl set image and blocks on kubectl rollout status until pods are Ready.

```
Jenkinsfile x
Jenkinsfile
1 // Jenkinsfile - SWE 645 Assignment 2 CI/CD pipeline. Written by Lizon Raj.
2 // On every push to GitHub, this pipeline builds a new Docker image tagged with the
3 // short git SHA, pushes it to Docker Hub, and rolls out the new image on EKS.
4
5 pipeline {
6   // Run on the built-in Jenkins controller node (the EC2 itself).
7   agent any
8
9   options {
10     // Fail the build if any single stage runs longer than 15 min - safety net for hung docker/kubectl calls.
11     timeout(time: 15, unit: 'MINUTES')
12     // Keep only the last 10 build histories to save disk on the t3.small.
13     buildDiscarder(logRotator(numToKeepStr: '10'))
14   }
15
16   environment {
17     IMAGE = 'lizonraj/lizon-swe645-hw2' // Docker Hub repo
18     DEPLOY = 'lizon-site' // Kubernetes Deployment name
19     CONTAINER = 'lizon-site' // Container name inside the pod (must match deployment.yaml)
20     NAMESPACE = 'default' // Kubernetes namespace
21     CLUSTER = 'lizon-swe645' // EKS cluster name (for logging)
22     REGION = 'us-east-1' // AWS region (for logging)
23   }
24
25   stages {
26     stage('Checkout') {
27       steps {
28         // Pulls source code from the git repo configured in the Jenkins job.
29         checkout scm
30       }
31     }
32
33     stage('Compute tag') {
34       steps {
35         script {
36           // Short 7-char commit SHA - becomes the immutable image tag for this build.
37           // Using a unique tag per commit is critical: it forces Kubernetes to
38           // recognize a spec change and roll out new pods (:latest alone would not).
39           env.TAG = sh(script: 'git rev-parse --short HEAD', returnStdout: true).trim()
40         }
41       }
42     }
43   }
44 }
```

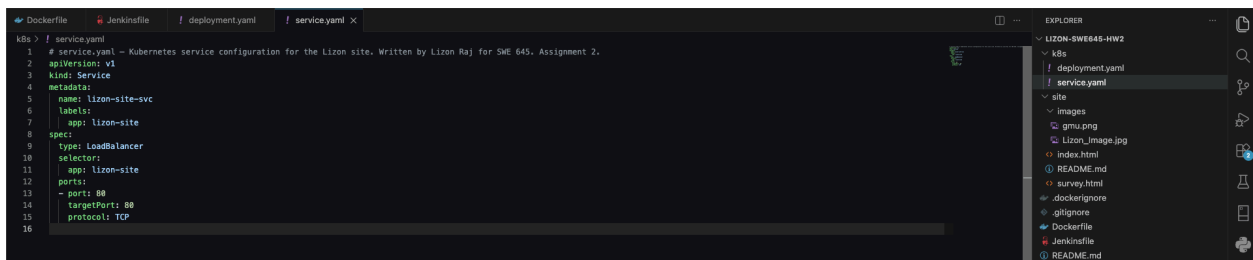
Deployment.yaml:

Kubernetes Deployment specification — 3 replicas of the nginx container, imagePullPolicy: IfNotPresent (safe with immutable SHA tags), liveness and readiness probes on / port 80 for self-healing, and modest CPU/memory requests and limits appropriate for a static file server. Labels and selectors both use app: lizon-site so the Service can locate the pods.



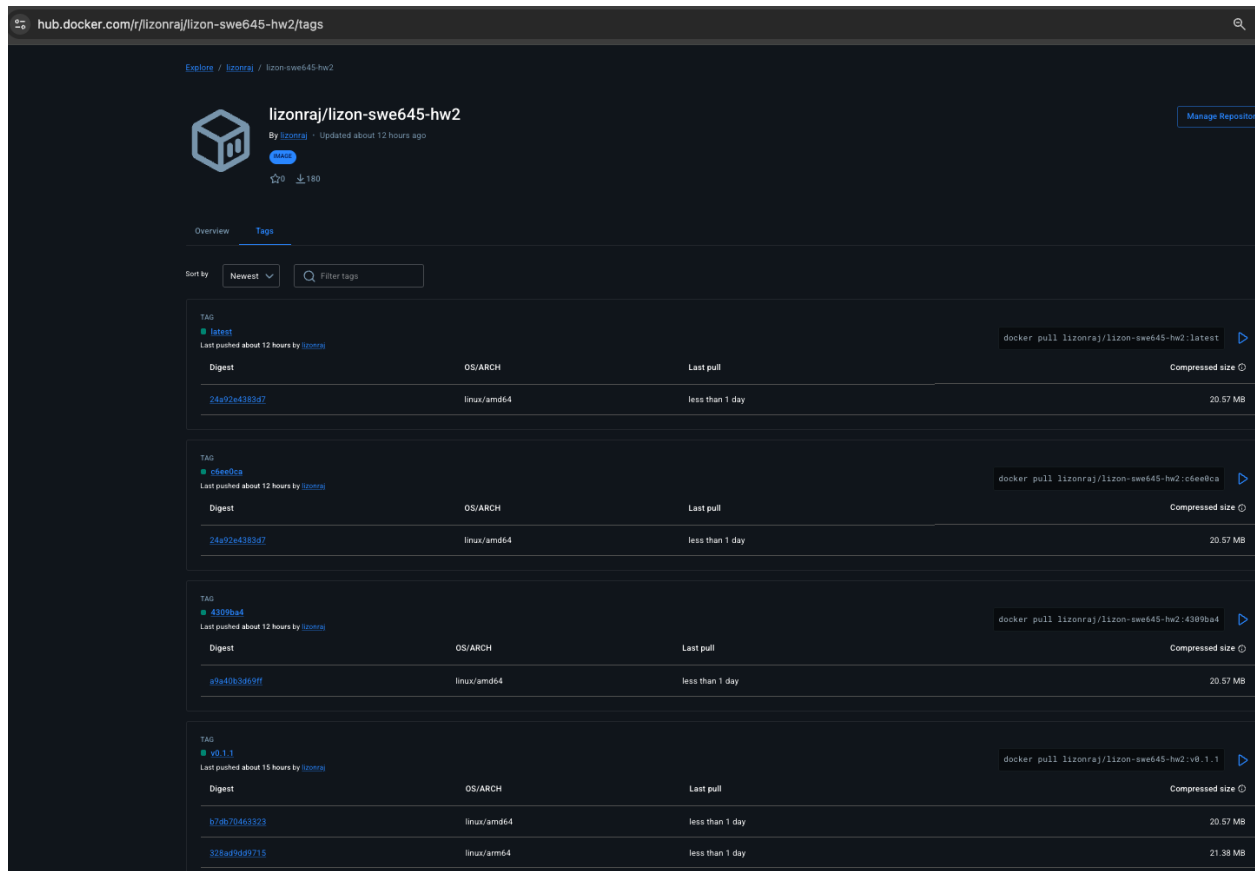
Service.yaml:

Kubernetes Service of type LoadBalancer on port 80 → targetPort 80 (TCP). AWS automatically provisions a Classic ELB with a public DNS hostname that routes traffic to the three Deployment pods.



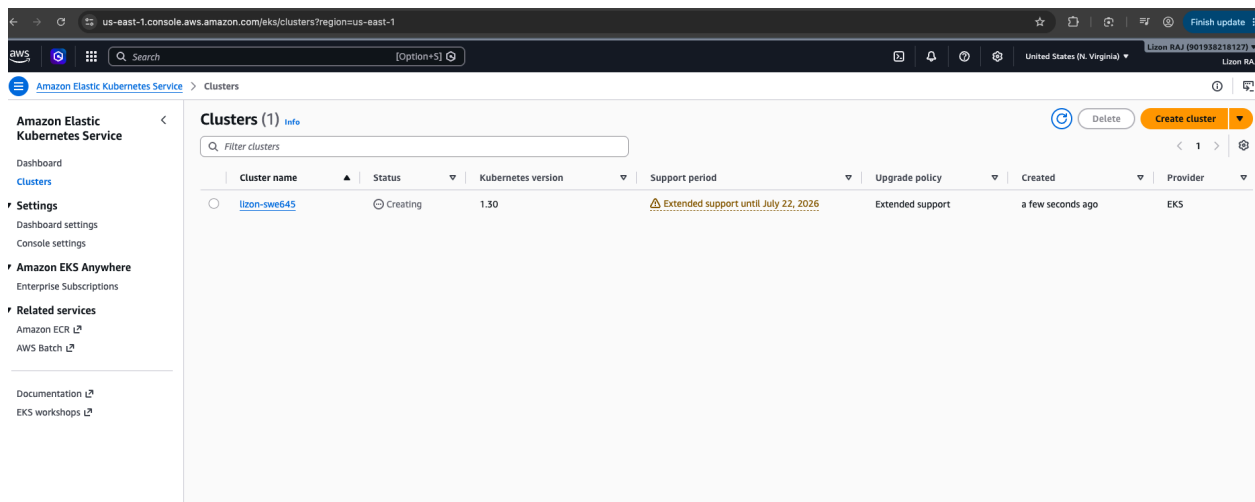
DockerHub:

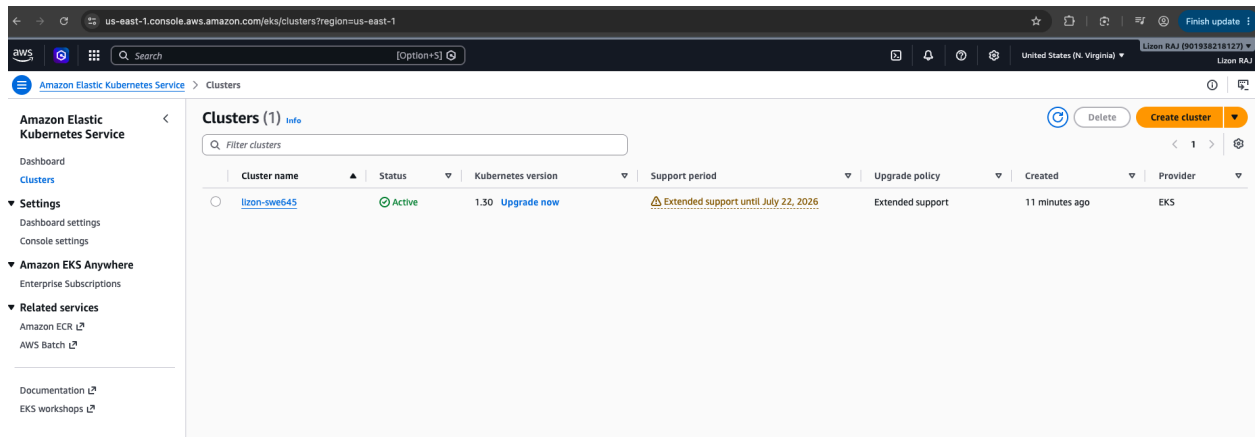
Public Docker Hub repository [lizonraj/lizon-swe645-hw2](https://hub.docker.com/r/lizonraj/lizon-swe645-hw2) with tags for latest, semantic versions (v0.1.0, v0.1.1), and per-commit short-SHA tags pushed by Jenkins on every build. The v0.1.1 and later tags are multi-architecture (linux/amd64 for EKS nodes, linux/arm64 for local Apple Silicon development).



Amazon EKS:

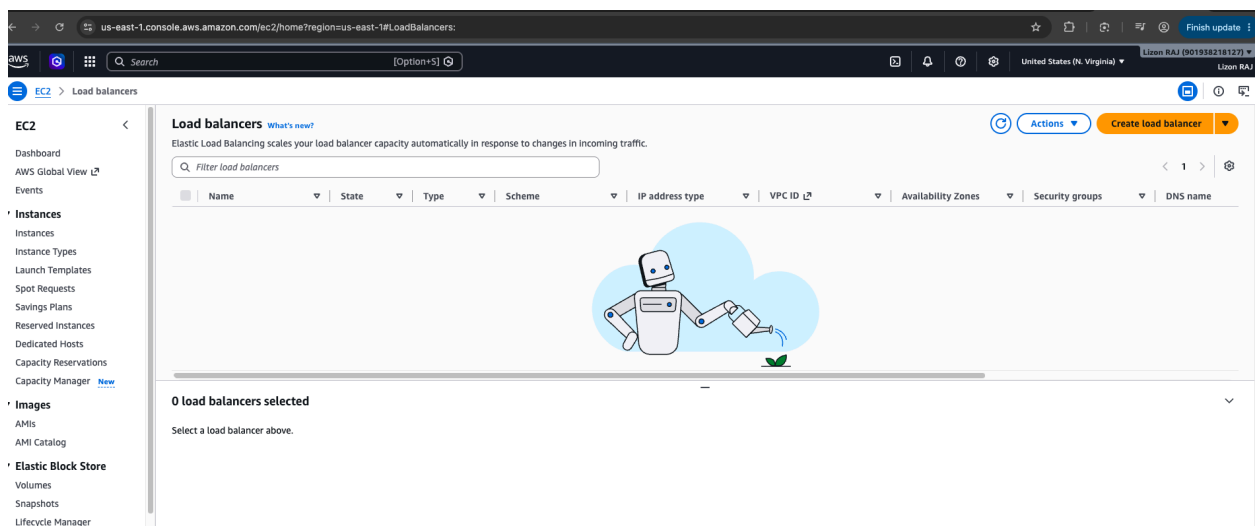
AWS EKS cluster lizon-swe645 in us-east-1, running Kubernetes 1.30 on a managed node group of two t3.medium worker nodes. Cluster provisioning was fully automated via eksctl create cluster.





Load Balancer:

AWS Classic Elastic Load Balancer automatically provisioned by the Kubernetes LoadBalancer Service. The public DNS hostname shown routes port 80 traffic to any healthy pod in the Deployment.



kubectl Deployment Images:

Output of `kubectl describe deployment lizon-site | grep Image` showing the currently deployed image tag matches a Jenkins-built short-SHA. This proves the CI/CD pipeline successfully updated the Deployment spec and Kubernetes rolled out new pods.

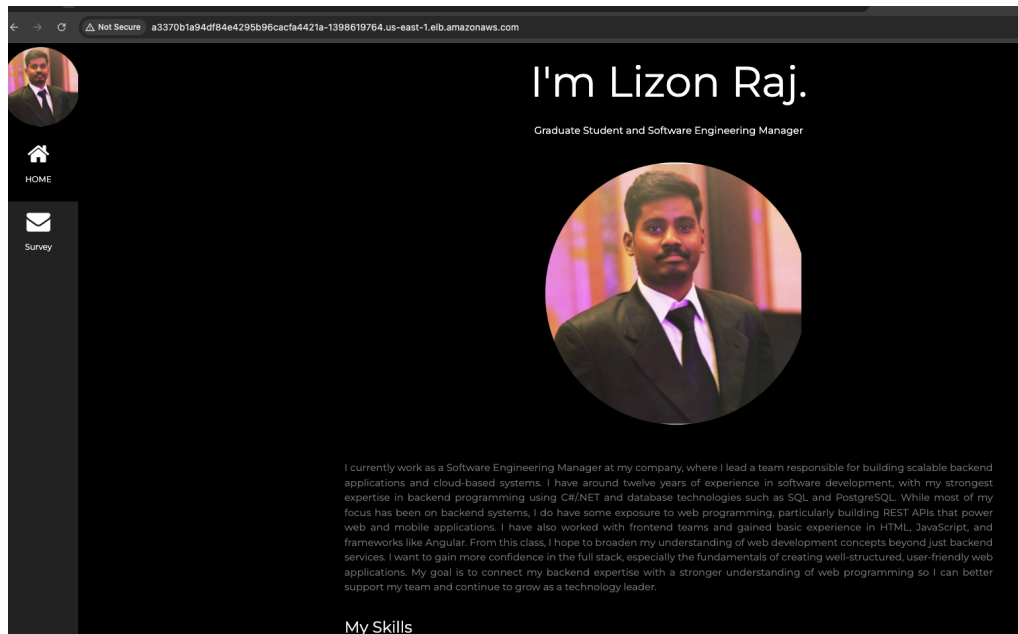
```
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl apply -f k8s/
deployment.apps/lizon-site created
service/lizon-site-svc created
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 %

lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
ip-192-168-20-64.ec2.internal        Ready    <none>    3m43s  v1.30.14-eks-8f14419
ip-192-168-39-102.ec2.internal        Ready    <none>    3m43s  v1.30.14-eks-8f14419
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl apply -f k8s/
deployment.apps/lizon-site created
service/lizon-site-svc created
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl get pods -l app=lizon-site -w
NAME                                READY    STATUS    RESTARTS    AGE
lizon-site-59ff6596d8-lr9js         1/1      Running    0            51s
lizon-site-59ff6596d8-rd5rl         1/1      Running    0            51s
lizon-site-59ff6596d8-rzrl8         1/1      Running    0            51s
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl get pods -o wide
NAME                                READY    STATUS    RESTARTS    AGE    IP              NODE                                NOMINATED NODE    READINESS GATES
lizon-site-59ff6596d8-lr9js         1/1      Running    0            2m7s   192.168.51.91   ip-192-168-39-102.ec2.internal      <none>             <none>
lizon-site-59ff6596d8-rd5rl         1/1      Running    0            2m7s   192.168.7.249   ip-192-168-20-64.ec2.internal       <none>             <none>
lizon-site-59ff6596d8-rzrl8         1/1      Running    0            2m7s   192.168.48.145  ip-192-168-39-102.ec2.internal      <none>             <none>
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % kubectl get svc lizon-site-svc -w
NAME      TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
lizon-site-svc  LoadBalancer  10.100.3.114   a3370b1a94df84e4295b96cacfa4421a-1398619764.us-east-1.elb.amazonaws.com  80:30280/TCP  2m24s
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % svc -w
zsh: command not found: svc
lizonraj@mac05-LQJ1WV9RXX lizon-swe645-hw2 % ELB=a3370b1a94df84e4295b96cacfa4421a-1398619764.us-east-1.elb.amazonaws.com
curl -I http://$ELB/
curl -I http://$ELB/survey.html
HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Sun, 19 Jul 2026 04:37:49 GMT
Content-Type: text/html
Content-Length: 4934
Last-Modified: Sun, 12 Jul 2026 11:42:20 GMT
Connection: keep-alive
ETag: "6a537d9c-1346"
Accept-Ranges: bytes

HTTP/1.1 200 OK
Server: nginx/1.27.5
Date: Sun, 19 Jul 2026 04:37:49 GMT
Content-Type: text/html
Content-Length: 8574
Last-Modified: Sun, 12 Jul 2026 11:42:38 GMT
Connection: keep-alive
ETag: "6a537dae-217e"
Accept-Ranges: bytes
```

Live Running Site:

Browser hits to the LoadBalancer URL serving index.html and survey.html — the original Assignment 1 pages now served from Kubernetes pods. The public ELB hostname is visible in the URL bar.

A screenshot of a "Prospective Student Survey" form. The form is titled "Prospective Student Survey:" in a green, sans-serif font. Below the title are several input fields for personal information: "First Name:", "Last Name:", "Street Address:", "City:", "State:", "Zip:", "Telephone:", "E-mail:", "URL:", "Date of Survey:", and "High School Graduation Month:". Each field has a placeholder text or a default value. The form is set against a dark background with a light-colored sidebar on the left containing "HOME" and "Survey" buttons. A logo with the letters "GM" is visible at the top of the form area.

Self-Healing Proof:

Terminal capture of `kubectl delete pod <name>` followed immediately by `kubectl get pods`, showing a replacement pod (different name suffix) automatically created within seconds. This satisfies the assignment's self-healing requirement — the Deployment controller continuously reconciles the desired replica count.


```
lizonraj@macOS-LQJ1WV9RXX lizon-swe645-hw2 % kubectl delete pod lizon-site-59ff6596d8-lr9js
pod "lizon-site-59ff6596d8-lr9js" deleted from default namespace
lizonraj@macOS-LQJ1WV9RXX lizon-swe645-hw2 % kubectl get pods -l app=lizon-site -w
NAME                                READY   STATUS    RESTARTS   AGE
lizon-site-59ff6596d8-rd5rl        1/1     Running   0           10m
lizon-site-59ff6596d8-rzrl8        1/1     Running   0           10m
lizon-site-59ff6596d8-zbg59        1/1     Running   0           9s
```

Jenkins:

The Jenkins dashboard with the lizon-swe645-pipeline job configured. Jenkins is running on an Amazon Linux 2023 t3.small EC2 instance in us-east-1, installed with Java 21 (required by Jenkins 2.568), Docker, and kubectl. An IAM instance profile (JenkinsEC2Role) grants EKS cluster-admin via an access entry, so kubectl from Jenkins works with no hardcoded AWS credentials.

a. Jenkins setup:

```

us-east-1-console.amazonaws.com | ec2-instance-connection:ssh home/regionus-east-1/console#type=standard&instanceid=i-b0b222d9b64d7956a408a0user=ec2-user&sshPort=22&addressfamily=ipv4
AWS [Search] [Option:5] [United States (N. Virginia)] [Lizson RAJ (P01958218127)] [Lizson RAJ]
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]> 17c3963061e406e80282252096c55d
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]> this may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.052+0000 [ide=40] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.094+0000 [ide=30] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal system[1]: Started jenkins.service - Jenkins Continuous Integration Server.
[ec2-user@ip-172-31-27-11 ~]$ sudo systemctl status jenkins --no-pager
jenkins.service - Jenkins Continuous Integration Server
Loaded: loaded (/usr/lib/systemd/system; jenkins.service; enabled; preset: disabled)
Active: active (running) since Sun 2026-07-19 05:29:27 UTC; 19s ago
Main PID: 29611 (java)
Tasks: 50 (limit: 2210)
Memory: 559.1M
CPU: 21.268s
CGroup: /system.slice/jenkins.service
└─29611 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:22 ip-172-31-27-11.ec2.internal jenkins[29611]: [LF]>
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.052+0000 [ide=40] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.094+0000 [ide=30] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal system[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.486+0000 [ide=56] INFO hudson.DownloadServiceDownloadable#load: Obtained the updated data file for hudson-maveninstaller
Jul 19 05:29:27 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:27.408+0000 [ide=56] INFO hudson.util.Retrier#start: Performed the action check updates server successful, the attempt #1
Jul 19 05:29:32 ip-172-31-27-11.ec2.internal jenkins[29611]: 2026-07-19 05:29:32.156+0000 [ide=78] WARNING h.n.DiskSpaceMonitorDescriptor#markModeOnFileOnUpdate: Making Built-In Nod.k of disk space
[ec2-user@ip-172-31-27-11 ~]$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
17c3963061e406e80282252096c55d
[ec2-user@ip-172-31-27-11 ~]$ sudo systemctl restart jenkins
[ec2-user@ip-172-31-27-11 ~]$ sudo -u jenkins aws eks update-kubeconfig --name lizon-swe645 --region us-east-1
Added new context arn:aws:eks:us-east-1:901938218127:cluster/lizon-swe645 to /var/lib/jenkins/.kube/config
[ec2-user@ip-172-31-27-11 ~]$ sudo -u jenkins kubectl get pods -l app=lizon-site
NAME READY STATUS RESTARTS AGE
lizon-site-59ff6596d8-rd5r1 1/1 Running 0 69m
lizon-site-59ff6596d8-rd5r2 1/1 Running 0 69m
lizon-site-59ff6596d8-rd5r3 1/1 Running 0 59m
[ec2-user@ip-172-31-27-11 ~]$

```

b. Jenkins running pipeline:

Stage View of a successful pipeline run showing all five stages green: Checkout, Compute tag, Docker build, Docker push, Deploy to EKS. Post-stage output confirms Deployed lizonraj/lizon-swe645-hw2:<sha> to EKS cluster lizon-swe645.

The screenshot shows the Jenkins web interface. The top navigation bar includes the Jenkins logo, the name of the pipeline 'lizon-swe645-pipeline', and the build number '#7'. The left sidebar contains various navigation options: Status, Changes, Console Output (selected), Delete build '#7', Git Build Data, Pipeline Overview, Edit build information, Restart from Stage, Replay, Pipeline Steps, Workspaces, and Previous Build. The main console area displays the following output:

```
Started by user Lizon
Obtained Jenkinsfile from git https://github.com/Lizonraj/lizon-swe645-hw2.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/lizon-swe645-pipeline
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
No credentials specified
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/lizon-swe645-pipeline/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/Lizonraj/lizon-swe645-hw2.git # timeout=10
Fetching upstream changes from https://github.com/Lizonraj/lizon-swe645-hw2.git
> git --version # timeout=10
> git --version # 'git version 2.50.1'
> git fetch --tags --force --progress -- https://github.com/Lizonraj/lizon-swe645-hw2.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision c6ee0ca53286e5e8bd46d58d52f9ad8067791b41 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f c6ee0ca53286e5e8bd46d58d52f9ad8067791b41 # timeout=10
Commit message: "test: trigger CI/CD via webhook"
> git rev-list --no-walk c6ee0ca53286e5e8bd46d58d52f9ad8067791b41 # timeout=10
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] timeout
Timeout set to expire in 15 min
[Pipeline] {
[Pipeline] stage
[Pipeline] {
```

Below the console output, there is a navigation bar with a back arrow, a forward arrow, a refresh icon, a 'Not Secure' warning, and the URL '54.196.145.70:8080/manage/credentials/'.



Jenkins | Manage Jenkins | Credentials

Credentials



dockerhub-creds lizonraj/*****

System - Global - Docker Hub push credentials

Stores scoped to Jenkins

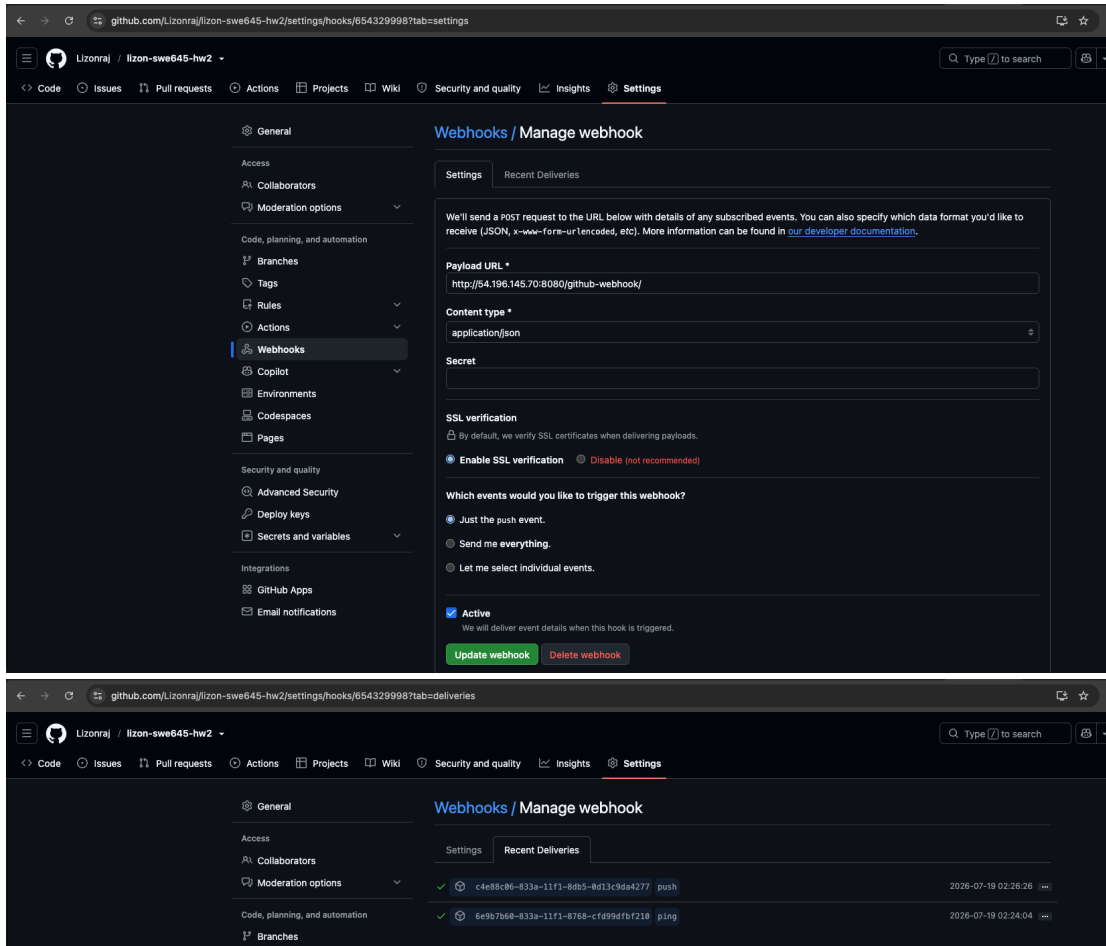


System

Domains: Global

Github Webhook:

The webhook configured on the GitHub repository pointing at `http://<jenkins-ip>:8080/github-webhook/`. The Recent Deliveries section shows successful push-event deliveries (green checkmarks), confirming that every git push to main reaches Jenkins and triggers a pipeline build automatically.



EC2 Instances:

The lizon-jenkins EC2 instance hosting the Jenkins controller. Instance type t3.small (2 vCPU, 2 GB RAM), Amazon Linux 2023, 20 GB gp3 root volume. Attached to the JenkinsEC2Profile IAM instance profile for keyless EKS access.

The screenshot shows the AWS Management Console 'Instances' page. The table below lists the instances:

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
lizon-swe645-ng-58e4bf32-node	i-0e6fc1eb087ec2169	Running	t3.medium	3/5 checks passed	us-east-1c	ec2-100-48-82-125.co...	100.48.82.125	-
lizon-swe645-ng-58e4bf32-node	i-06d0d065f4bde4c49	Running	t3.medium	3/5 checks passed	us-east-1f	ec2-3-239-8-252.comp...	3.239.8.252	-
lizon-jenkins	i-0b0222d9b64d7556a	Running	t3.small	3/5 checks passed	us-east-1c	ec2-54-196-145-70.co...	54.196.145.70	-

Security Group and Inbound rules:

Security group jenkins-sg restricting inbound traffic to two rules: SSH (port 22) from the developer's public IP only, and Jenkins HTTP (port 8080) from 0.0.0.0/0 so GitHub's webhook servers can reach the pipeline endpoint. Jenkins' built-in authentication protects the UI itself.

The screenshot displays the AWS Management Console interface for an EC2 instance. The left sidebar shows navigation options like EC2, Dashboard, and various instance management tools. The main content area is titled 'I-0e6fc1eb087ec2169' and includes tabs for Details, Status and alarms, Monitoring, Security, Networking, Storage, and Tags. The 'Security' tab is active, showing 'Security details' with the IAM role 'eksctl-lizon-swe645-nodegroup-ng-5-NodeInstanceRole-8DtcA2R55vQP' and the IAM role 'eksctl-lizon-swe645-cluster/SubnetPublicUSEAST1C'. It also lists 'Security groups' as 'sg-0a1d7cd30ac510030 (eks-cluster-sg-lizon-swe645-1080178489)'. Below this, the 'Inbound rules' section shows a table with three rules: 'sgr-095750b127f56bde' (SSH), 'sgr-0b8818aae34dd7677' (Jenkins HTTP), and 'sgr-0a015dff8ac06f067' (All traffic). The 'Outbound rules' section shows a single rule 'sgr-07c6f4a447b3afeb2' (All traffic). The 'Launch time' is 'Sun Jul 19 2026 00:25:27 GMT-0400 (Eastern Daylight Time)'.

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#InstanceDetails:instanceId=I-0e6fc1eb087ec2169

EC2 > Instances > I-0e6fc1eb087ec2169

Dashboard
AWS Global View
Events

▼ Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

▼ Images
AMIs
AMI Catalog

▼ Elastic Block Store
Volumes
Snapshots
Lifecycle Manager

▼ Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces

▼ Load Balancing
Load Balancers
Target Groups
Trust Stores

Details Status and alarms Monitoring **Security** Networking Storage Tags

▼ Security details

IAM role
eksctl-lizon-swe645-nodegroup-ng-5-NodeInstanceRole-8DtcA2R55vQP

Subnet ID
subnet-05a385855ae549d9c (eksctl-lizon-swe645-cluster/SubnetPublicUSEAST1C)

Instance ARN
arn:aws:ec2:us-east-1:901938218127:instance/I-0e6fc1eb087ec2169

Auto Scaling Group name
eks-ng-58e40f52-92c7bc4f-48ff-6d64-f72c-cf03604f1b20

Managed
false

Owner ID
901938218127

Launch time
Sun Jul 19 2026 00:25:27 GMT-0400 (Eastern Daylight Time)

▼ Inbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sgr-095750b127f56bde	All	All	sg-0a1d7cd30ac510030	eks-cluster-sg-lizon-swe645-1080178...	-
-	sgr-0b8818aae34dd7677	All	All	sg-022b877f266b644e2	eks-cluster-sg-lizon-swe645-1080178...	-
-	sgr-0a015dff8ac06f067	All	All	sg-0e32fe5b44bf8486d	eks-cluster-sg-lizon-swe645-1080178...	Allow unmanaged nodes to comm

▼ Outbound rules

Filter rules

Name	Security group rule ID	Port range	Protocol	Destination	Security groups	Description
-	sgr-07c6f4a447b3afeb2	All	All	0.0.0.0/0	eks-cluster-sg-lizon-swe645-1080178...	-